THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 01

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** an exothermic reaction	Heat is released
2	**(d)** sodium hydroxide	Fully ionised
3	**(a)** Correct	Definition
4	**(d)** 2	2, 8, 2
5	**(c)** gold	1 g gold → 2 km wire
6	**(b)** C_nH_{2n-2}	Triple bond
7	**(c)** ethanoic acid	5–8%
8	**(d)** night	Store CO_2 as acid
9	**(c)** cerebellum	Precision of voluntary actions
10	**(a)** Student B is correct	Not 100% accurate
11	**(c)** vasectomy	Surgical method
12	**(a)** 3	Rryy/RRyy
13	**(a)** provides information for one protein	Codes protein
14	**(b)** Student A is correct	Sign convention
15	**(c)** hypermetropia	Far-sighted
16	**(b)** rheostat	Variable resistor
17	**(b)** 12 Ω	$R \propto l$
18	**(a)** Oersted	1820
19	**(a)** decomposers	Recycling
20	**(d)** kitchen waste	Biodegradable

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	It forms a thin, tough layer of aluminium oxide that protects it from further reaction.	2
22	It has fewer H_3O^+ ions per unit volume.	2
23	Autotrophs make food from simple inorganic substances (e.g. green plants); heterotrophs depend on other organisms for food (e.g. animals).	2
24	Their pancreas does not secrete enough insulin, so blood sugar remains high.	2
25	Produces eggs (ova) and secretes hormones (oestrogen).	2
26	False — they do not change the DNA of germ cells, so they are not inherited.	2
27	(i) concave (ii) convex	1+1
28	Concentric circles in a horizontal plane; anticlockwise when viewed from above.	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	(i) $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ (ii) $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2\,NaCl$ (iii) $2\,Mg + O_2 \longrightarrow 2\,MgO$	1+1+1
30	Electrolysis of molten Al_2O_3: cathode $Al^{3+} + 3\,e^- \longrightarrow Al$; anode $2\,O^{2-} \longrightarrow O_2 + 4\,e^-$	1+1+1
31	Auxin: cell elongation at shoot tip; gibberellin: stem growth; cytokinin: cell division, found in fruits and seeds	1+1+1
32	Many disease-free plants from one plant quickly; used for orchids, ornamentals; no seeds needed	3
33	Retina converts light to electrical signals; optic nerve sends them to the brain; rods sense dim light, cones sense colour	3
34	Metal body connected to earth via green wire; leakage current flows to earth, keeping the body at zero potential, preventing shock	3

35	Distilled water has almost no ions; rain water contains dissolved acids/salts which provide ions.	3
	OR: HCl/H_2SO_4 solutions glow a bulb in a circuit (ions present); glucose/alcohol solutions do not — they do not give H^+.	2+1
36	Increase in UV radiation causing skin cancer, cataract, weakening of immune system, damage to plants and aquatic life	3
	OR: Energy decreases at each level; less energy available supports fewer organisms	3

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) convex mirror (ii) image is diminished so objects appear farther (iii) $f = +1$ m (iv) $\frac{1}{v} = 1 + \frac{1}{10} = \frac{11}{10} \Rightarrow v = +0.91$ m (v) no, virtual	1+1+1+1+1
	OR: (i) (a) at C, real, inverted, same size (b) beyond C, real, inverted, magnified (c) at infinity, real, highly magnified (ii) ray parallel to axis passes through F; ray through C retraces	3+2
38	(i) long hydrocarbon tail (hydrophobic) and ionic head (hydrophilic) (ii) micelles trap oil inside, washed away by water (iii) forms insoluble scum with Ca^{2+}/Mg^{2+} (iv) effective in hard water	1+2+1+1
	OR: (i) (a) ethanoic acid turns blue litmus red; ethanol no change (b) acid gives CO_2; alcohol no reaction (c) ethanol pleasant; acid vinegar smell (ii) CH_3CH_2OH; CH_3COOH	3+2
39	(i) $R = 5/2.5 = 2\ \Omega$; $10/n = 2 \Rightarrow n = 5$ (ii) two in series (8 Ω) in parallel with two in series (8 Ω) gives 4 Ω	3+2
	OR: (i) a continuous closed path for current; standard symbols (ii) so that it draws negligible current and does not change the potential difference being measured	4+1
40	(i) xylem vessels and tracheids; root cells take ions, water enters by osmosis; transpiration pull (ii) pressure created by water entering roots pushing water up (iii) at night when transpiration is low	3+1+1
	OR: (i) synthesis of food from CO_2 and water using sunlight and chlorophyll; $6\,CO_2 + 12\,H_2O \longrightarrow C_6H_{12}O_6 + 6\,O_2 + 6\,H_2O$ (ii) gas exchange, transpiration (iii) stored as starch	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER